

2.1

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$$1) (3x+2y)dx + (2x+y)dy = 0$$

$$\frac{\partial M}{\partial y} = 2 \quad \frac{\partial N}{\partial x} = 2$$

$$\text{Since, } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Hence, the equation is exact

$$3xdx + 2ydx + 2xdy + ydy = 0$$

$$\Rightarrow \frac{3}{2}d(x^2) + \frac{1}{2}d(y^2) + 2d(xy) = 0$$

$$\Rightarrow d\left(\frac{3}{2}x^2 + \frac{1}{2}y^2 + 2xy\right) = 0$$

By integrating,

$$\frac{3}{2}x^2 + \frac{1}{2}y^2 + 2xy = C \quad \underline{\text{Ans.}}$$

$$2) (y^2+3)dx + (2xy-4)dy = 0$$

$$\frac{\partial M}{\partial y} = 2y, \quad \frac{\partial N}{\partial x} = 2y$$

So, exact

$$\Rightarrow y^2dx + 3dx + 2xydy - 4dy = 0$$

$$\Rightarrow d(xy^2) + d(3x) - d(4y) = 0$$

$$\Rightarrow d(xy^2 + 3x - 4y) = 0$$

By integrating,

$$xy^2 + 3x - 4y = C \quad \underline{\text{Ans.}}$$

$$3) (2xy+1)dx + (x^2+4y)dy = 0$$

$$\frac{\partial M}{\partial y} = 2x$$

$$\frac{\partial N}{\partial x} = 2x$$

so, exact

$$\Rightarrow 2xydx + dx + x^2dy + 4ydy = 0$$

$$\Rightarrow d(x^2y) + d(x) + d(2y^2) = 0$$

$$\Rightarrow d(x^2y + x + 2y^2) = 0$$

By integrating,

$$x^2y + x + 2y^2 = C \quad \underline{\text{Ans.}}$$

$$4) (3x^2y + 2)dx - (x^3 + y)dy = 0$$

$$\frac{\partial M}{\partial y} = 3x^2 \quad \frac{\partial N}{\partial x} = -3x^2$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ So, the equation is not exact. Ans.

$$5) (6xy + 2y^2 - 5)dx + (3x^2 + 4xy - 6)dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 6x + 4y \quad ; \text{ so exact}$$

$$6xydx + 2y^2dx - 5dx + 3x^2dy + 4xydy - 6dy = 0$$

$$\Rightarrow d(-5x) + d(-6y) + d(3x^2y) + d(2xy^2) = 0$$

$$\Rightarrow \text{By integrating, } \Rightarrow d(-5x - 6y + 3x^2y + 2xy^2) = 0$$

$$\Rightarrow 3x^2y + 2xy^2 - 5x - 6y = C \quad \underline{\underline{\text{Ans.}}}$$

$$6) (\theta^2 + 1) \cos r dr + 2\theta \sin r d\theta = 0$$

$$\Rightarrow (\theta^2 \cos r + \cos r) dr + (2\theta \sin r) d\theta = 0$$

$$\frac{\partial M}{\partial \theta} = 2\theta \cos r \quad \frac{\partial N}{\partial r} = 2\theta \cos r$$

So, exact

$$\theta^2 \cos r dr + \cos r dr + 2\theta \sin r d\theta = 0$$

$$\Rightarrow d(\theta^2 \sin r) + d(\sin r) = 0$$

$$\Rightarrow d(\theta^2 \sin r + \sin r) = 0$$

$\Rightarrow$ By integrating,

$$\theta^2 \sin r + \sin r = C$$

Ans.

$$7) (y \sec^2 x + \sec x \tan x) dx + (\tan x + 2y) dy = 0$$

$$\frac{\partial M}{\partial y} = \sec^2 x \quad \frac{\partial N}{\partial x} = \sec^2 x$$

so, exact

$$y \sec^2 x dx + \sec x \tan x dx + \tan x dy + 2y dy = 0$$

$$\Rightarrow d(y \tan x) + d(\sec x) + d(y^2) = 0$$

$$\Rightarrow d(y \tan x + \sec x + y^2) = 0$$

By integrating,

$$y \tan x + \sec x + y^2 = c \quad \underline{\text{Ans.}}$$

$$8) \left(\frac{x}{y^2} + x \right) dx + \left(\frac{x^2}{y^3} + y \right) dy = 0$$

$$\frac{\partial M}{\partial y} = -\frac{2x}{y^3} \quad \frac{\partial N}{\partial x} = \frac{2x}{y^3}$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ The equation is not exact Ans.

$$9) \left(\frac{2s-1}{t} \right) ds + \left(\frac{s-s^2}{t^2} \right) dt = 0$$

$$\frac{\partial M}{\partial t} = \frac{-2s+1}{t^2} \quad \frac{\partial N}{\partial s} = \frac{1-2s}{t^2}$$

$$\frac{\partial M}{\partial t} = \frac{\partial N}{\partial s}; \text{ so the equation is exact}$$

$$\frac{2s}{t} ds - \frac{1}{t} ds + \frac{s}{t^2} dt - \frac{s^2}{t^2} dt = 0$$

$$\Rightarrow d\left(\frac{s^2}{t}\right) + d\left(-\frac{s}{t}\right) = 0$$

$$\Rightarrow d\left(\frac{s^2}{t} - \frac{s}{t}\right) = 0$$

By integrating,

$$\frac{s^2}{t} - \frac{s}{t} = c \quad \underline{\text{Ans.}}$$

$$10) \frac{2y^{3/2}+1}{x^{1/2}} dx + (3x^{1/2}y^{1/2}-1) dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{3y^{1/2}}{x^{1/2}} \quad \frac{\partial N}{\partial x} = \frac{3}{2} x^{-1/2} y^{1/2} = \frac{3y^{1/2}}{2x^{1/2}}$$

$$\text{Since, } \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Hence, the equation is not exact. Ans.

$$11) (2xy-3)dx + (x^2+4y)dy=0 \quad ; \quad y(1)=2$$

$$\Rightarrow 2xydx - 3dx + x^2dy + 4ydy = 0$$

$$\Rightarrow d(x^2y) + d(-3x) + d(2y^2) = 0$$

$$\Rightarrow d(x^2y - 3x + 2y^2) = 0$$

By integrating,

$$x^2y - 3x + 2y^2 = C \dots \text{general solution}$$

$$\Rightarrow (1)^2 \cdot 2 - 3 \cdot 1 + 2(2)^2 = C$$

$$\Rightarrow C = 7$$

$$\therefore x^2y - 3x + 2y^2 = 7 \dots \text{particular solution}$$

$$12) (3x^2y^2 - y^3 + 2x)dx + (2x^3y - 3xy^2 + 1)dy = 0 \quad ; \quad y(-2) = 1$$

$$\Rightarrow 3x^2y^2dx - y^3dx + 2xdx + 2x^3ydy - 3xy^2dy + dy = 0$$

$$\Rightarrow d(x^2+y + x^3y^2 - xy^3) = 0$$

By integrating,

$$x^2+y + x^3y^2 - xy^3 = C$$

$$\Rightarrow (-2)^2 + 1 + (-2)^3 \cdot 1^2 - (-2) \cdot 1 = C$$

$$\Rightarrow C = -1$$

$$\text{So, the equation is: } x^3y^2 + x^2 - xy^3 + y = -1 \quad \underline{\text{Ans.}}$$

$$13) (2y \sin x \cos x + y^2 \sin x) dx + (\sin^2 x - 2y \cos x) dy = 0 ; \quad y(0) = 3$$

$$\Rightarrow 2y \sin x \cos x dx + y^2 \sin x dx + \sin^2 x dy - 2y \cos x dy = 0$$

$$\Rightarrow d(-y^2 \cos x) + d(y \sin^2 x) = 0$$

$$\Rightarrow d(y \sin^2 x - y^2 \cos x) = 0$$

By integrating,

$$y \sin^2 x - y^2 \cos x = c$$

Now,

$$3(\sin(0))^2 - 3^2 \cos(0) = c$$

$$\Rightarrow c = -9$$

So, the equation is: $y \sin^2 x - y^2 \cos x = -9$ Ans.

$$14) (ye^x + 2e^x + y^2) dx + (e^x + 2xy) dy = 0 ; \quad y(0) = 6$$

$$\Rightarrow ye^x dx + 2e^x dx + y^2 dx + e^x dy + 2xy dy = 0$$

$$\Rightarrow d(xy^2) + d(2e^x) + d(ye^x) = 0$$

$$\Rightarrow d(xy^2 + 2e^x + ye^x) = 0$$

By integrating,

$$xy^2 + 2e^x + ye^x = c$$

$$\Rightarrow 0 \cdot y^2 + 2e^0 + ye^0 = c$$

$$\Rightarrow c = 8$$

So, the equation is: $xy^2 + 2e^x + ye^x = 8$ Ans.

$$15) \left(\frac{3-y}{x^2} \right) dx + \left(\frac{y^2-2x}{xy^2} \right) dy = 0 \quad ; \quad y(-1) = 2$$

$$\Rightarrow \frac{3}{x^2} dx - \frac{y}{x^2} dx + \frac{y^2}{xy^2} dy - \frac{2x}{xy^2} dy = 0$$

$$\Rightarrow d\left(-3 \cdot \frac{1}{x}\right) + d\left(\frac{y}{x}\right) + d\left(\frac{2}{y}\right) = 0$$

$$\Rightarrow d\left(-\frac{3}{x} + \frac{y}{x} + \frac{2}{y}\right) = 0$$

By integrating,

$$-\frac{3}{x} + \frac{y}{x} + \frac{2}{y} = C$$

Now,

$$-\frac{3}{-1} + \frac{2}{-1} + \frac{2}{2} = C$$

$$\Rightarrow C = 2$$

So, the equation is: $-\frac{3}{x} + \frac{y}{x} + \frac{2}{y} = 2$ Ans.

$$16) \frac{1+8xy^{\frac{2}{3}}}{x^{\frac{2}{3}}y^{\frac{1}{3}}} dx + \frac{2x^{\frac{4}{3}}y^{\frac{2}{3}} - x^{\frac{1}{3}}}{y^{\frac{4}{3}}} dy = 0 \quad ; \quad y(1) = 8$$

$$\Rightarrow d\left(6x^{\frac{4}{3}}y^{\frac{1}{3}}\right) + d\left(3 \frac{x^{\frac{1}{3}}}{y^{\frac{1}{3}}}\right) = 0$$

By integrating;

$$6x^{\frac{4}{3}}y^{\frac{1}{3}} + 3 \frac{x^{\frac{1}{3}}}{y^{\frac{1}{3}}} = C$$

Now,

$$6 \cdot 1^{\frac{4}{3}} \cdot 8^{\frac{1}{3}} + 3 \frac{1^{\frac{1}{3}}}{8^{\frac{1}{3}}} = C$$

$$\Rightarrow C = \frac{27}{2}$$

So, the equation is: $12x^{\frac{4}{3}}y^{\frac{1}{3}} + 6 \frac{x^{\frac{1}{3}}}{y^{\frac{1}{3}}} = 27$ Ans.

$$1) 4xy dx + (x^2+1) dy = 0$$

$$\Rightarrow \frac{4x dx}{x^2+1} + \frac{dy}{y} = 0$$

By integrating,

$$\Rightarrow \int \frac{4x dx}{x^2+1} + \int \frac{dy}{y} = c$$

$$\Rightarrow 2 \int \frac{2x dx}{x^2+1} + \int \frac{1}{y} dy = c$$

$$\Rightarrow 2 \log(x^2+1) + \log y = \log c$$

$$\Rightarrow (x^2+1)^2 y = c$$

$$\Rightarrow x^4 y + 2x^2 y + y = c \quad \underline{\underline{\text{Ans.}}}$$

$$2) (xy+2x+y+2) dx + (x^2+2x) dy = 0$$

$$\Rightarrow (x+1)(y+2) dx + (x^2+2x) dy = 0$$

$$\Rightarrow \frac{x+1}{x^2+2x} dx + \frac{dy}{y+2} = 0$$

By integrating,

$$\Rightarrow \frac{1}{2} \int \frac{2x+2}{x^2+2x} dx + \int \frac{dy}{y+2} = c$$

$$\Rightarrow \frac{1}{2} \log(x^2+2x) + \log(y+2) = \log c$$

$$\Rightarrow \log(x^2+2x)^{1/2} + \log(y+2) = \log c$$

$$\Rightarrow (x^2+2x)^{1/2} (y+2) = c \quad \underline{\underline{\text{Ans.}}}$$

$$3) 2r(s^2+1) dr + (r^4+1) ds = 0$$

$$\Rightarrow \frac{2r}{r^4+1} dr + \frac{ds}{s^2+1} = 0$$

$$\Rightarrow \frac{dr^2}{(r^2)^2+1} + \frac{ds}{s^2+1} = 0$$

By integrating,

$$\tan^{-1} r^2 + \tan^{-1} s = c \quad \underline{\text{Ans.}}$$

$$5) \tan \theta dr + 2r d\theta = 0$$

$$\Rightarrow \frac{dr}{2r} + \frac{d\theta}{\tan \theta} = 0$$

By integrating,

$$\frac{1}{2} \int \frac{dr}{r} + \int \frac{d\theta}{\tan \theta} = c$$

$$\Rightarrow \frac{1}{2} \log r + \int \frac{\cos \theta}{\sin \theta} d\theta = c$$

$$\Rightarrow \frac{1}{2} \log r + \log \sin \theta = \log c$$

$$\Rightarrow \sqrt{r} \sin \theta = c \quad \underline{\text{Ans.}}$$

$$4) \operatorname{cosec} y dx + \sec x dy = 0$$

$$\Rightarrow \frac{dx}{\sec x} + \frac{dy}{\operatorname{cosec} y} = 0$$

By integrating,

$$\int \cos x dx + \int \sin y dy = c$$

$$\Rightarrow \sin x - \cos y = c \quad \underline{\text{Ans.}}$$

$$6) (e^v + 1) \cos u du + e^v (\sin u + 1) dv = 0$$

$$\Rightarrow \frac{\cos u}{\sin u + 1} du + \frac{e^v}{e^v + 1} dv = 0$$

By integrating,

$$\log(\sin u + 1) + \log(e^v + 1) = \log c$$

$$\Rightarrow (\sin u + 1)(e^v + 1) = c \quad \underline{\text{Ans.}}$$

$$7) (x+4)(y^2+1)dx + y(x^2+3x+2)dy = 0$$

$$\Rightarrow \frac{x+4}{x^2+3x+2} dx + \frac{y dy}{y^2+1} = 0$$

By integrating,

$$\int \frac{x+4}{(x+2)(x+1)} dx + \frac{1}{2} \int \frac{2y}{y^2+1} dy = c \dots (i)$$

NOW,

$$\frac{x+4}{(x+2)(x+1)} = \frac{A}{x+2} + \frac{B}{x+1}$$

$$\Rightarrow x+4 = A(x+1) + B(x+2)$$

$$\text{if } x=-1 : 3 = B$$

$$\text{if } x=-2 : 2 = -A \therefore A = -2$$

$$\begin{aligned} \text{So, } \int \frac{x+4}{(x+2)(x+1)} &= - \int \frac{2}{x+2} dx + 3 \int \frac{dx}{x+1} \\ &= -2 \ln(x+2) + 3 \ln(x+1) \end{aligned}$$

putting these values to equation (i),

$$\ln(x+2)^{-2} + \ln(x+1)^3 + \ln(y^2+1)^{1/2} = \ln c$$

$$\Rightarrow (x+2)^{-2} (x+1)^3 (y^2+1)^{1/2} = c$$

Ans.

$$8) (x+y)dx - xdy = 0$$

$$\Rightarrow (x+y)dx = xdy$$

$$\Rightarrow \frac{dy}{dx} = \frac{x+y}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x+vx}{x} = 1+v$$

$$\Rightarrow x \frac{dv}{dx} = 1$$

$$\Rightarrow dv = \frac{dx}{x}$$

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + \frac{dv}{dx} x$$

By integrating,

$$v = \ln x + C$$

$$\Rightarrow \frac{y}{x} = \ln x + C$$

$$\Rightarrow \frac{y}{x} - \ln x = C \quad \underline{\text{Ans.}}$$

$$9) (2xy + 3y^2) dx - (2xy + x^2) dy = 0$$

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2xy + 3y^2}{2xy + x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{2x^2v + 3v^2x^2}{2x^2v + x^2} = \frac{2v + 3v^2}{2v + 1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2v + 3v^2 - 2v^2 - v}{2v + 1} = \frac{v^2 + v}{2v + 1}$$

$$\Rightarrow \frac{2v + 1}{v^2 + v} dv = \frac{dx}{x}$$

By integrating,

$$\ln(v^2 + v) - \ln x = \ln C$$

$$\Rightarrow \frac{x}{v^2 + v} = C$$

$$\Rightarrow \frac{x}{\frac{y^2}{x^2} + \frac{y}{x}} = C$$

$$\Rightarrow \frac{x^3}{y^2 + yx} = C \quad \underline{\text{Ans.}}$$

$$10) v^3 du + (u^3 - uv^2) dv = 0$$

$$u = vs$$

$$\Rightarrow \frac{du}{dv} = - \frac{u^3 - uv^2}{v^3}$$

$$\Rightarrow \frac{du}{dv} = s + v \frac{ds}{dv}$$

$$\Rightarrow s + v \frac{ds}{dv} = - \frac{v^3 s^3 - v^3 s}{v^3} = -s^3 + s$$

$$\Rightarrow v + \frac{ds}{dv} = -s^3 \Rightarrow -\frac{ds}{ds^3} = \frac{dv}{v}$$

$$\Rightarrow s^{-3} ds + \frac{dv}{v} = 0$$

By integrating,

$$\frac{s^{-2}}{-2} + \ln v = c \Rightarrow \frac{1}{-2s^2} + \ln v = c$$

$$\Rightarrow \frac{1}{-2 \cdot \frac{u^2}{v^2}} + \ln v = c \Rightarrow \frac{v^2}{-2u^2} + \ln v = c \quad \underline{\text{Ans.}}$$

$$11) (x \tan \frac{y}{x} + y) dx - x dy = 0$$

$$y = vx \Rightarrow v = y/x$$

$$\Rightarrow \frac{dy}{dx} = \frac{x \tan \frac{y}{x} + y}{x}$$

$$\frac{dy}{dx} = v + x \cdot \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x \tan v + vx}{x} = \tan v + v$$

$$\Rightarrow x \frac{dv}{dx} = \tan v \Rightarrow \frac{dv}{\tan v} = \frac{dx}{x} \Rightarrow \frac{\cos v}{\sin v} dv = \frac{dx}{x}$$

By integrating,

$$\ln(\sin v) = \ln x + \ln c$$

$$\Rightarrow \frac{\sin v}{x} = c \Rightarrow \frac{\sin y/x}{x} = c \quad \underline{\text{Ans.}}$$

$$12) (2s^2 + 2st + t^2) ds + (s^2 + 2st - t^2) dt = 0$$

$$\Rightarrow \frac{ds}{dt} = - \frac{s^2 + 2st - t^2}{2s^2 + 2st + t^2}$$

$$s = rt$$

$$\Rightarrow \frac{ds}{dt} = r + t \frac{dr}{dt}$$

$$\Rightarrow r + t \frac{dr}{dt} = - \frac{r^2 t^2 + 2rt^3 - t^2}{2r^2 t^2 + 2rt^2 + t^2} = - \frac{r^2 + 2r - 1}{2r^2 + 2r + 1}$$

$$\Rightarrow t \frac{dr}{dt} = - \frac{r^2 + 2r - 1 + 2r^3 + 2r^2 + r}{2r^2 + 2r + 1}$$

$$\Rightarrow t \frac{dr}{dt} = - \frac{2r^3 + 3r^2 + 3r - 1}{2r^2 + 2r + 1}$$

$$\Rightarrow \frac{2r^2 + 2r + 1}{2r^3 + 3r^2 + 3r - 1} dr = - \frac{dt}{t}$$

By integrating,

$$\frac{1}{3} \int \frac{6r^2 + 6r + 3}{2r^3 + 3r^2 + 3r - 1} dr = - \int \frac{dt}{t}$$

$$\Rightarrow \frac{1}{3} \ln(2r^3 + 3r^2 + 3r - 1) + \ln t = \ln c$$

$$\Rightarrow \frac{1}{3} \ln\left(2\frac{s^3}{t^3} + 3\frac{s^2}{t^2} + 3\frac{s}{t} - 1\right) + \ln t = \ln c$$

$$\Rightarrow \left(\frac{2s^3}{t^3} + \frac{3s^2}{t^2} + \frac{3s}{t} - 1\right)^{1/3} = c \quad \underline{\underline{\text{Ans.}}}$$

$$13) (x^3 + y^2 \sqrt{x^2 + y^2}) dx - xy \sqrt{x^2 + y^2} dy = 0$$

$$y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^3 + y^2 \sqrt{x^2 + y^2}}{xy \sqrt{x^2 + y^2}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^3 + v^2 x^2 \sqrt{x^2 + v^2 x^2}}{x^2 v \sqrt{x^2 + v^2 x^2}} = \frac{x^3 + v^2 x^3 \sqrt{1 + v^2}}{x^3 v \sqrt{1 + v^2}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1 + v^2 \sqrt{1 + v^2}}{v \sqrt{1 + v^2}}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2 \sqrt{1 + v^2} - v^2 \sqrt{1 + v^2}}{v \sqrt{1 + v^2}} = \frac{1}{v \sqrt{1 + v^2}}$$

$$\Rightarrow v \sqrt{1 + v^2} dv = \frac{dx}{x}$$

By integrating,

$$\int v \sqrt{1 + v^2} dv = \ln x = c$$

$$\Rightarrow \int \frac{1}{2} \sqrt{z} dz - \ln x = c$$

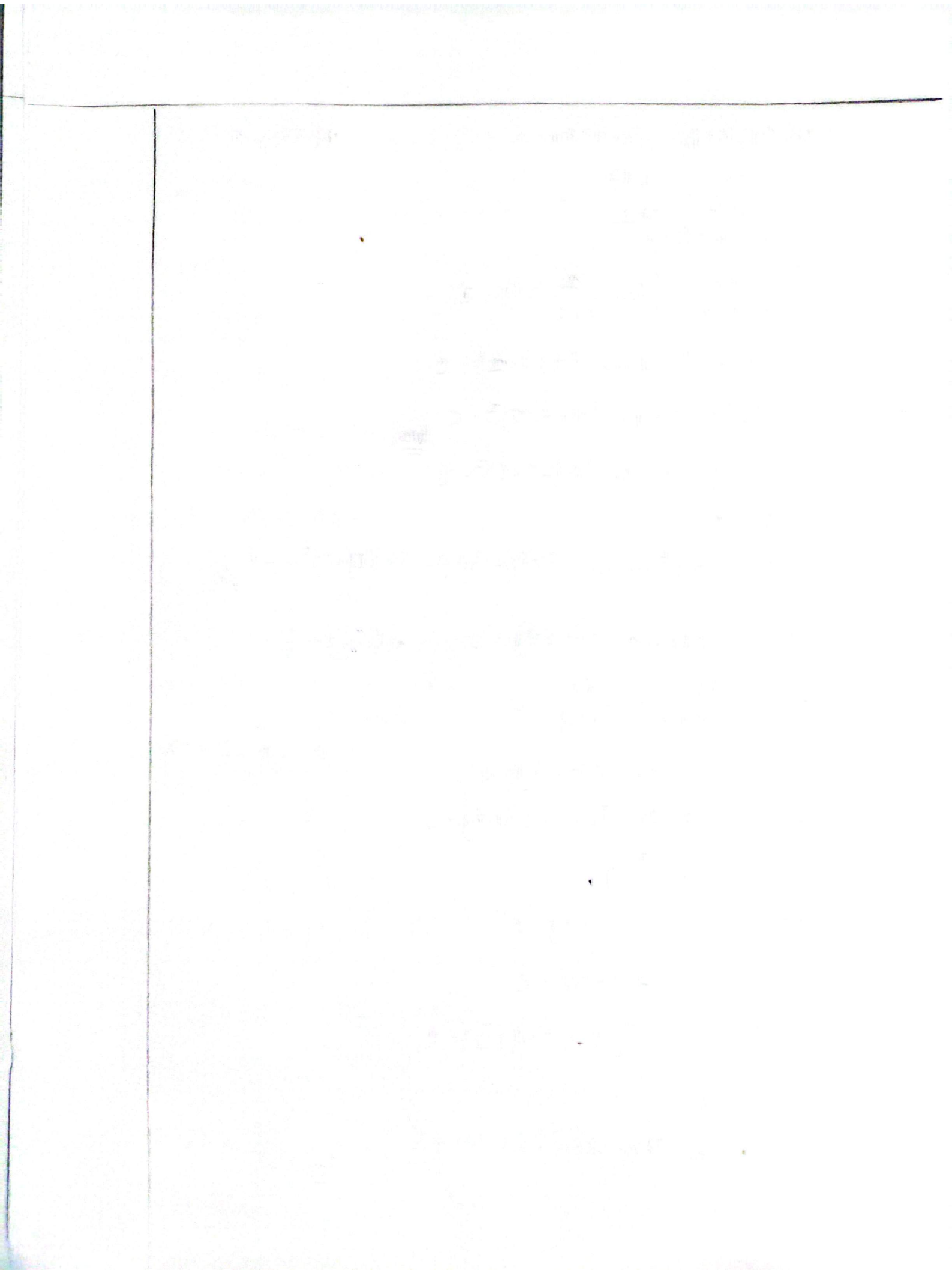
$$\Rightarrow \int \frac{1}{2} \frac{z^{3/2}}{3/2} - \ln x = c \Rightarrow \frac{1}{3} (1 + v^2)^{3/2} - \ln x = c$$

$$\Rightarrow \frac{1}{3} \left(1 + \frac{y^2}{x^2}\right)^{3/2} - \ln x = c \quad \underline{\underline{\text{Ans.}}}$$

$$\text{let, } 1 + v^2 = z$$

$$\Rightarrow 2v dv = dz$$

$$\Rightarrow v dv = \frac{dz}{2}$$



$$15) (y+2)dx + y(x+4)dy = 0$$

$$y(-3) = -1$$

$$\Rightarrow \frac{dx}{x+4} + \frac{y dy}{y+2} = 0$$

By integrating,

$$\Rightarrow \int \frac{dx}{x+4} + \int \left(1 - \frac{2}{y+2}\right) dy = C$$

$$\Rightarrow \ln(x+4) + y - 2 \ln(y+2) = C$$

$$\Rightarrow \ln(x+4) + y - \ln(y+2)^2 = C$$

$$\Rightarrow \ln(-3+4) - 1 - \ln(-1+2)^2 = C$$

$$\Rightarrow C = -1$$

$$\therefore \text{The solution is } \ln(x+4) + y - \ln(y+2)^2 = -1 \quad \underline{\underline{\text{Ans.}}}$$

$$16) 8\cos^2 y dx + \sec^2 x dy = 0$$

$$y\left(\frac{\pi}{2}\right) = \frac{\pi}{4}$$

$$\Rightarrow 8 \frac{dx}{\sec^2 x} + \frac{dy}{\cos^2 y} = 0$$

$$\Rightarrow 8 \sin^2 x dx + \sec^2 y dy = 0$$

$$1 - \cos 2x = 2 \sin^2 x$$

$$\Rightarrow 4(1 - \cos 2x) dx + \sec^2 y dy = 0$$

By integrating,

$$4x - 4 \frac{\sin 2x}{2} + \tan y = C$$

$$\Rightarrow 4x - 2 \sin 2x + \tan y = C$$

$$\Rightarrow \frac{\pi}{3} - 2 \sin\left(\frac{\pi}{6}\right) + \tan\left(\frac{\pi}{4}\right) = C$$

$$\Rightarrow C = \frac{\pi}{3}$$

$$\therefore \text{Solution: } 4x - 2 \sin 2x + \tan y = \frac{\pi}{3}$$

Ans.

$$17) (3x+8)(y^2+4) dx - 4y(x^2+5x+6) dy = 0 \quad y(1) = 2$$

$$\Rightarrow \frac{3x+8}{x^2+5x+6} dx - \frac{4y}{y^2+4} dy = 0 \dots (i)$$

Now,

$$\frac{3x+8}{x^2+5x+6} = \frac{A}{x+3} + \frac{B}{x+2}$$

$$\Rightarrow 3x+8 = A(x+2) + B(x+3)$$

$$\text{if } x = -2 : B = 2$$

$$\text{if } x = -3 : A = 1$$

Integrating eq (i)

$$\int \frac{1}{x+3} dx + \int \frac{2}{x+2} dx - 2 \int \frac{2y}{y^2+4} dy = C$$

$$\Rightarrow \ln(x+3) + \ln(x+2)^2 - \ln(y^2+4)^2 = \ln e$$

$$\Rightarrow \frac{(x+3)(x+2)^2}{(y^2+4)^2} = e$$

$$\Rightarrow \frac{(1+3)(1+2)^2}{((2)^2+4)^2} = e$$

$$\Rightarrow e = \frac{9}{16}$$

$$\therefore \text{The equation is: } \frac{(x+3)(x+2)^2}{(y^2+4)^2} = \frac{9}{16} \quad \underline{\underline{Ans.}}$$

$$18) (x^2+3y^2) dx - 2xy dy = 0 \quad y(2) = 6$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2+3y^2}{2xy}$$

$$y = vx \\ \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^2+3v^2x^2}{2x^2v} = \frac{1+3v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+3v^2-2v^2}{2v} = \frac{v^2+1}{2v}$$

$$\Rightarrow \ln \left(5 \frac{y^2}{x^2} + 1 \right)^{1/5} - \ln(x) = c$$

By integrating,

$$\ln(v^2+1) - \ln x = c \Rightarrow \ln \left(\frac{y^2}{x^2} + 1 \right) - \ln(x) = \ln c$$

$$\Rightarrow \ln \left(\frac{y^2}{x^3} + \frac{1}{x} \right) = \ln c$$

$$\Rightarrow c = \frac{y^2}{x^3} + \frac{1}{x} = 5$$

So, the solution is: $\frac{y^2}{x^3} + \frac{1}{x} = 5$ Ans.

$$10) (2x - 5y) dx + (4x - y) dy = 0 \quad y(1) = 4$$

$$\Rightarrow \frac{dy}{dx} = \frac{5y - 2x}{4x - y}$$

let, $y = vx$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{5vx - 2x}{4x - vx} = \frac{5v - 2}{4 - v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{5v - 2 - 4v + v^2}{4 - v} = \frac{v^2 + v - 2}{4 - v}$$

$$\Rightarrow \frac{(4 - v) dv}{v^2 + v - 2} = \frac{dx}{x}$$

$$20) (3x^2 + 9xy + 5y^2)dx - (6x^2 + 4xy)dy = 0 \quad y(2) = -6$$

$$\frac{dy}{dx} = \frac{3x^2 + 9xy + 5y^2}{6x^2 + 4xy}$$

$$y = vx \\ \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{3x^2 + 9x^2v + 5x^2v^2}{6x^2 + 4x^2v} = \frac{3 + 9v + 5v^2}{6 + 4v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{3 + 9v + 5v^2 - 6v - 4v^2}{6 + 4v} = \frac{v^2 + 3v + 3}{6 + 4v}$$

$$\Rightarrow \frac{4v + 6}{v^2 + 3v + 3} dv = \frac{dx}{x}$$

By integrating,

$$2 \int \frac{2v + 3}{v^2 + 3v + 3} dv = \ln x + c$$

$$\Rightarrow 2 \ln(v^2 + 3v + 3) - \ln(x) = \ln c$$

$$\Rightarrow \ln\left(\frac{y^2}{x^2} + \frac{3y}{x} + 3\right)^2 - \ln(x) = \ln c$$

$$\Rightarrow \frac{y^2}{x^3} + \frac{3y}{x^2} + \frac{3}{x} = c$$

$$\Rightarrow \frac{(-6)^2}{2^3} + \frac{3(-6)}{2^2} + \frac{3}{2} = c$$

$$\Rightarrow c = 3/2$$

So, the eq is: $\frac{y^2}{x^3} + \frac{3y}{x^2} + \frac{3}{x} = 3/2$ Ans.

2.3

2.3

$$1) \frac{dy}{dx} + \frac{3y}{x} = 6x^2$$

$$IF = e^{\int P dx} = e^{\int \frac{3}{x} dx} = x^3$$

$$\text{Soln: } y IF = \int IF Q dx$$

$$\Rightarrow y \cdot x^3 = \int x^3 \cdot 6x^2 dx$$

$$\Rightarrow y = x^3 + Cx^{-3} \quad \underline{\underline{\text{Ans.}}}$$

$$2) x^4 \frac{dy}{dx} + 2x^3 y = 1$$

$$\Rightarrow \frac{dy}{dx} + \frac{2}{x} y = \frac{1}{x^4}$$

$$IF = e^{\int P dx} = x^2$$

$$\text{Soln: } y IF = \int IF Q dx$$

$$\Rightarrow y = -\frac{1}{x^3} + \frac{C}{x^2} \quad \underline{\underline{\text{Ans.}}}$$

$$3) \frac{dy}{dx} + 3y = 3x^2 e^{-3x}$$

$$IF = e^{\int P dx} = e^{3x}$$

$$\text{Soln: } e^{3x} y = \int e^{3x} \cdot 3x^2 e^{-3x} dx$$

$$\Rightarrow y = (x^3 + C) e^{-3x} \quad \underline{\underline{\text{Ans.}}}$$

$$4) \frac{dy}{dx} + 4xy = 8x$$

$$IF = e^{\int 4x dx} = e^{2x^2}$$

$$\text{Soln: } e^{2x^2} y = \int e^{2x^2} \cdot 8x dx$$

$$\Rightarrow e^{2x^2} y = 2(e^{2x^2} + C)$$

$$\Rightarrow y = 2 + Ce^{-2x^2} \quad \underline{\underline{\text{Ans.}}}$$

$$5) \frac{dx}{dt} + \frac{x}{t^2} = \frac{1}{t^2}$$

$$IF = e^{\int P dt} = e^{-1/t}$$

$$\text{Soln: } e^{-1/t} x = \int e^{-1/t} \cdot \frac{1}{t^2} dt$$

$$\Rightarrow x = 1 + Ce^{1/t} \quad \underline{\underline{\text{Ans.}}}$$

$$6) (u^2+1) \frac{dv}{du} + uvv = 3u$$

$$\Rightarrow \frac{dv}{du} + \frac{uvv}{u^2+1} = \frac{3u}{u^2+1}$$

$$IF = e^{\int P du} = e^{2 \ln(u^2+1)} = (u^2+1)^2$$

$$\text{Soln: } (u^2+1)^2 v = \int \frac{3u}{u^2+1} \cdot (u^2+1)^2 du$$

$$= \int 3u^3 + 3u du$$

$$\Rightarrow 4(u^2+1)v = 3u^4 + 6u^2 + C \quad \underline{\underline{\text{Ans.}}}$$

$$7) x \frac{dy}{dx} + \frac{2x+1}{x+1} y = x-1$$

$$\Rightarrow \frac{dy}{dx} + \frac{2x+1}{x^2+x} y = \frac{x-1}{x}$$

$$IF = e^{\int P dx} = x^2+x$$

$$\text{Soln: } (x^2+x)y = \int \frac{x-1}{x} \cdot x(x+1) dx$$

$$\Rightarrow (x^2+x)y = \int x^2 - 1 dx$$

$$\Rightarrow 3(x^2+x)y = x^3 - 3x + C$$

$$\underline{\underline{\text{Ans.}}}$$

$$8) (x^2+x-2) \frac{dy}{dx} + 3(x+1)y = x-1$$

$$\Rightarrow \frac{dy}{dx} + \frac{3(x+1)y}{x^2+2x-x-2} = \frac{x-1}{(x-1)(x+2)}$$

$$\Rightarrow \frac{dy}{dx} + \frac{3(x+1)y}{(x-1)(x+2)} = \frac{1}{x+2}$$

$$P = \frac{3(x+1)}{(x-1)(x+2)} \quad Q = \frac{1}{x+2}$$

$$I.F. = e^{\int P dx} = e^{\int \frac{3(x+1)}{(x-1)(x+2)} dx}$$

Now,

$$\frac{x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

$$(x+1) = A(x+2) + B(x-1)$$

$$\text{if } x = -2, \text{ then } B = -1$$

$$\text{if } x = 1, \text{ then } A = \frac{2}{3}$$

$$\text{Hence, } \frac{x+1}{(x-1)(x+2)} = \frac{2}{3(x-1)} - \frac{1}{x+2}$$

$$I.F. = e^{\ln(x+2) + 2 \ln(x-1)} = (x+2)(x-1)^2$$

$$\text{Soln: } (x+2)(x-1)^2 y = \int (x-1)^2 (x+2) \cdot \frac{1}{x+2} dx$$

$$\Rightarrow (x-1)^2 (x+2) y = \frac{(x-1)^3}{3} + C/3$$

Ans.

$$9) x dy + (xy + y - 1) dx = 0$$

$$\Rightarrow \frac{dy}{dx} + y \left(1 + \frac{1}{x}\right) = \frac{1}{x}$$

$$I.F. = e^{\int P dx} = x e^x$$

$$\text{Soln: } x \cdot e^x y = \int \frac{1}{x} x e^x dx$$

$$\Rightarrow x y e^x = e^x + C$$

$$\Rightarrow xy = 1 + C e^{-x}$$

Ans.

$$10) y dx + (xy^2 + x - y) dy = 0$$

$$\Rightarrow \frac{dx}{dy} + \frac{xy^2 + x - y}{y} = 0$$

$$\Rightarrow \frac{dx}{dy} + x \left\{ y + \frac{1}{y} \right\} = 1$$

$$IF = e^{\int P dy} = y e^{y^2/2}$$

$$\text{Soln: } y x e^{y^2/2} = \int y e^{y^2/2} \cdot dy$$

$$\Rightarrow xy = 1 + c e^{y^2/2} \quad \underline{\text{Ans.}}$$

$$11) \frac{dr}{d\theta} + r \tan \theta = \cos \theta$$

$$IF = e^{\int P d\theta} = \sec \theta$$

$$\text{Soln: } r \sec \theta = \int \sec \theta \cdot \cos \theta d\theta$$

$$\Rightarrow r = \cos \theta (\theta + c) \quad \underline{\text{Ans.}}$$

$$12) \cos \theta dr + (r \sin \theta - \cos^3 \theta) d\theta = 0$$

$$\Rightarrow \frac{dr}{d\theta} = -r \tan \theta + \cos^3 \theta$$

$$\Rightarrow \frac{dr}{d\theta} + r \tan \theta = \cos^3 \theta$$

$$IF = e^{\int \tan \theta d\theta} = \sec \theta$$

$$\text{Soln: } r \sec \theta = \int \cos \theta \cdot \sec \theta \cdot \frac{1}{2} 2 \cos^2 \theta \cdot d\theta$$

$$\Rightarrow r \sec \theta = \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$\Rightarrow 2r = (\theta + \sin \theta \cos \theta + c) \cos \theta \quad \underline{\text{Ans.}}$$

$$16) x \frac{dy}{dx} + y = -2x^6 y^4$$

$$\Rightarrow y^{-4} \frac{dy}{dx} + \frac{1}{x} y^{-3} = -2x^5$$

$$\Rightarrow -\frac{1}{3} \frac{dv}{dx} + \frac{v}{x} = -2x^5$$

$$\Rightarrow \frac{dv}{dx} - \frac{3v}{x} = 6x^5$$

$$I.F. = e^{\int P dx} = \frac{1}{x^3}$$

$$\text{Soln: } \frac{v}{x^3} = \int (x^5 \times \frac{1}{x^3}) dx$$

$$\Rightarrow \frac{1}{x^3 y^3} = 2x^3 + C \quad \underline{\text{Ans.}}$$

let,

$$V = y^{1-4} = y^{-3}$$

$$\frac{dv}{dx} = -3y^{-4} \frac{dy}{dx}$$

$$17) dy + (4y - 8y^{-3})x dx = 0$$

$$\Rightarrow \frac{dy}{dx} = (8y^{-3} - 4y)x$$

$$\Rightarrow \frac{dy}{dx} + 4yx = 8xy^{-3}$$

$$\Rightarrow y^3 \frac{dy}{dx} + 4y^4 x = 8x$$

$$\Rightarrow \frac{1}{4} \frac{dv}{dx} + 4vx = 8x$$

$$\frac{dv}{dx} + 16xv = 32x$$

$$I.F. = e^{\int P dx} = e^{8x^2}$$

$$\text{Soln: } v \cdot e^{8x^2} = \int 32x \cdot e^{8x^2} dx$$

$$\Rightarrow y^4 = 2(1 + ce^{-8x^2}) \quad \underline{\text{Ans.}}$$

let,

$$v = y^{1-(-3)} = y^4$$

$$\frac{dv}{dx} = 4y^3 \frac{dy}{dx}$$

$$13) (\cos^2 x - y \cos x) dx - (1 + \sin x) dy = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos^2 x - y \cos x}{1 + \sin x}$$

$$\Rightarrow \frac{dy}{dx} + \frac{\cos x}{1 + \sin x} y = \frac{\cos^2 x}{1 + \sin x}$$

$$IF = e^{\int P dx} = 1 + \sin x$$

$$\text{Soln: } (1 + \sin x) y = \int (1 + \sin x) \cdot \frac{\cos^2 x}{1 + \sin x} dx$$

$$\Rightarrow (1 + \sin x) y = \frac{1}{2} (\theta + \sin \theta \cos \theta + C) \quad \underline{\underline{\text{Ans.}}}$$

$$14) (y \sin 2x - \cos x) dx + (1 + \sin^2 x) dy = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos x - y \sin 2x}{1 + \sin^2 x}$$

$$\Rightarrow \frac{dy}{dx} + y \frac{2 \sin x \cos x}{1 + \sin^2 x} = \frac{\cos x}{1 + \sin^2 x}$$

$$IF = e^{\int P dx} = 1 + \sin^2 x$$

$$\text{Soln: } (1 + \sin^2 x) y = \int \frac{\cos x}{1 + \sin^2 x} (1 + \sin^2 x) dx$$

$$\Rightarrow (1 + \sin^2 x) y = \sin x + C \quad \underline{\underline{\text{Ans.}}}$$

$$15) \frac{dy}{dx} - \frac{y}{x} = -\frac{y^2}{x}$$

$$\Rightarrow -\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = \frac{1}{x}$$

$$\Rightarrow \frac{dv}{dx} + \frac{v}{x} = \frac{1}{x}$$

$$IF = e^{\int P dx} = x$$

$$\text{Soln: } x \cdot v = \int \frac{1}{x} \times x dx$$

$$\Rightarrow x \cdot \frac{1}{y} = x + C \quad \underline{\underline{\text{Ans.}}}$$

let,

$$v = y^{1-2} = y^{-1}$$

$$\frac{dv}{dx} = -\frac{1}{y^2} = \frac{dy}{dx}$$

$$18) \frac{dx}{dt} + \frac{t+1}{2t} x = \frac{t+1}{xt}$$

$$\Rightarrow x \frac{dx}{dt} + \frac{t+1}{2t} x^2 = \frac{t+1}{t}$$

$$\Rightarrow \frac{dy}{dt} + \frac{t+1}{t} y = \frac{2(t+1)}{t}$$

$$IF = e^{\int P dt} = t \cdot e^t$$

$$\text{Soln: } y \cdot t \cdot e^t = \int \frac{2(t+1)}{t} \cdot t e^t dt$$

$$\Rightarrow x^2 \cdot t \cdot e^t = 2 + C \quad \underline{\text{Ans.}}$$

$$19) x \frac{dy}{dx} - 2y = 2x^4$$

$$y(2) = 8$$

$$IF = e^{\int P dx} = x^{-2}$$

$$\text{Soln: } yx^{-2} = \int 2x^3 \cdot x^{-2} dx$$

$$\Rightarrow yx^{-2} = 2(x^2 + C) \quad \text{..... (i)}$$

$$\Rightarrow \frac{1}{2} \times 8 \times \frac{1}{2} = 2^2 + C$$

$$\Rightarrow C = 1$$

$$\text{The soln is : } y = x^4 + 2x^2 \quad \underline{\text{Ans.}}$$

$$20) \frac{dy}{dx} + 3x^2 y = x^2$$

$$y(0) = 2$$

$$IF = e^{\int P dx} = e^{x^3}$$

$$\text{Soln: } ye^{x^3} = \int x^2 \cdot e^{x^3} dx = \frac{1}{3} = \frac{1}{3} \int 3x^2 e^{x^3} dx$$

$$\Rightarrow ye^{x^3} = \frac{1}{3} (e^{x^3} + C) \quad \text{..... (i)}$$

$$2 \times 1 = \frac{1}{3} (1 + C)$$

$$\Rightarrow C = 5$$

$$\therefore \text{exact solution is: } ye^{x^3} = \frac{1}{3} (e^{x^3} + 5) \quad \underline{\text{Ans.}}$$

let,

$$y = x^{1-(-1)} = x^2$$

$$\frac{dy}{dt} = 2x \frac{dx}{dt}$$

$$21) e^x [y - 3(e^x + 1)^2] dx + (e^x + 1) dy = 0$$

$$y(0) = 4$$

$$\Rightarrow \frac{dy}{dx} = \frac{e^x [3(e^x + 1)^2 - y]}{e^x + 1}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y e^x}{1 + e^x} = \frac{3(e^x + 1)^2 e^x}{1 + e^x}$$

$$I.F. = e^{\int P dx} = 1 + e^x$$

$$\text{Soln: } y(1 + e^x) = \int 3e^x (1 + e^x)(1 + e^x) dx$$

$$\Rightarrow y(1 + e^x) = (1 + e^x)^3 + C \dots \textcircled{i}$$

$$\Rightarrow 4(1 + 1) = (1 + 1)^3 + C$$

$$\Rightarrow C = 0$$

$$\text{exact solution is: } y = (1 + e^x)^2$$

$$22) 2x(y + 1) dx - (x^2 + 1) dy = 0 \quad y(0) = 4$$

$$\frac{dy}{dx} = \frac{2x(y + 1)}{1 + x^2}$$

$$I.F. = e^{\int P dx} = (1 + x^2)^{-1}$$

$$\text{Soln: } y(1 + x^2)^{-1} = \int \frac{2x}{(1 + x^2)^2} dx$$

$$\Rightarrow y(1 + x^2)^{-1} = -\frac{1}{1 + x^2} + C \dots \textcircled{i}$$

$$\Rightarrow 4 \times 1 = -1 + C$$

$$\Rightarrow C = 5$$

$$\therefore \text{exact solution is: } y(1 + x^2)^{-1} = -\frac{1}{1 + x^2} + 5$$

Ans.

$$23) \frac{dr}{d\theta} + r \tan \theta = \cos^2 \theta \quad r\left(\frac{\pi}{4}\right) = 1$$

$$IF = e^{\int p d\theta} = e^{\int \tan \theta d\theta} = \sec \theta$$

$$\text{Soln: } r \sec \theta = \int \sec \theta \cdot \cos^2 \theta d\theta$$

$$\Rightarrow r \sec \theta = \sin \theta + C$$

$$\Rightarrow C = \frac{1}{\sqrt{2}}$$

$$\therefore \text{exact solution is: } r \sec \theta = \sin \theta + \frac{1}{\sqrt{2}} \quad \underline{\underline{\text{Ans.}}}$$

$$24) \frac{dx}{dt} - x = \sin 2t \quad x(0) = 0$$

$$IF = e^{\int -1 dt} = e^{-t}$$

$$\text{Soln: } x e^{-t} = \int \sin 2t \cdot e^{-t} dt$$

Now,

$$I = \int \sin 2t \cdot e^{-t} dt$$

$$= -\frac{1}{2} \cos 2t \cdot e^{-t} - \frac{1}{2} \left[\frac{1}{2} \sin 2t \cdot e^{-t} - \frac{1}{2} \int e^{-t} \sin 2t dt \right]$$

$$\Rightarrow \frac{5}{4} I = -\frac{1}{4} \sin 2t \cdot e^{-t} - \frac{1}{2} \cos 2t \cdot e^{-t} + C$$

$$\Rightarrow 5 x e^{-t} = -\sin 2t \cdot e^{-t} - 2 \cos 2t \cdot e^{-t} + C \dots \textcircled{i}$$

$$\Rightarrow 0 = (-0 - 2 + C) \frac{1}{5} \Rightarrow C = \frac{2}{5}$$

$$\text{exact solution, } 5 x e^{-t} = -\sin 2t \cdot e^{-t} - 2 \cos 2t e^{-t} + \frac{2}{5} \quad \underline{\underline{\text{Ans.}}}$$

$$\underline{25.} \quad \frac{dy}{dx} + \frac{y}{2x} = \frac{x}{y^3}$$

$$\Rightarrow y^3 \frac{dy}{dx} + \frac{y^4}{2x} = x$$

$$\Rightarrow \frac{dv}{dx} + \frac{4v}{2x} = 4x$$

$$y(1) = 2 \quad \left| \begin{array}{l} \text{let,} \\ v = y^{1-(-3)} = y^4 \\ \frac{dv}{dx} = 4y^3 \frac{dy}{dx} \end{array} \right.$$

$$\text{IF} = x^2$$

$$\text{Soln: } vx^2 = \int 4x \cdot x^2 dx$$

$$\Rightarrow y^4 x^2 = x^4 + c \dots \textcircled{i}$$

$$\Rightarrow c = 1$$

$$\therefore \text{exact solution is } y^4 x^2 = x^4 + 1 \quad \underline{\underline{\text{Ans.}}}$$

$$\underline{26)} \quad x \frac{dy}{dx} + y = (xy)^{3/2}$$

$$\Rightarrow y^{-3/2} \frac{dy}{dx} + \frac{y^{-1/2}}{x} = x^{1/2}$$

$$\Rightarrow \frac{dv}{dx} - \frac{v}{2x} = \frac{x^{1/2}}{2}$$

$$y(1) = 4$$

$$\left| \begin{array}{l} \text{let,} \\ v = y^{1-3/2} = y^{-1/2} \\ \frac{dv}{dx} = -\frac{1}{2} y^{-3/2} \end{array} \right.$$

$$P = -\frac{1}{2x} \quad Q = x^{1/2}$$

$$\text{IF} = x^{-1/2}$$

$$\text{Soln: } vx^{-1/2} = \frac{1}{2} \int x^{1/2} \cdot x^{-1/2} dx$$

$$\Rightarrow \frac{1}{\sqrt{xy}} = -\frac{x}{2} + c \dots \textcircled{i}$$

$$\Rightarrow \cancel{c} \quad c = 1$$

$$\therefore \text{exact solution is: } \frac{1}{\sqrt{xy}} = -\frac{x}{2} + 1 \quad \underline{\underline{\text{Ans.}}}$$